

Sai Vijay Ragav

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EDUCATION

Rajalakshmi Institute of Technology

B.E. in Computer Science (Artificial Intelligence & Machine Learning)

Chennai, Tamil Nadu

2023 – 2027

Indian Institute of Technology

Diploma in Data Science

Chennai, Tamil Nadu

2024 – 2026

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Dart

Tech: Spring Boot, FastAPI, REST APIs, Microservices, LangChain, LangGraph

Databases/Tools: PostgreSQL, Firebase, Milvus, Git, Docker, Maven

Core: Data Structures & Algorithms, Object Oriented Programming, DBMS, Operating Systems, Computer Networks

EXPERIENCE

AI/ML Intern

Oct 2025 – Dec 2025

Sprelia

Remote

- Developed a multimodal RAG pipeline using **LangChain** and **Unstructured** to ingest, parse, and index over **10,000 patent documents**, adding metadata to improve downstream retrieval and analysis.
- Designed a **hybrid search** system backed by **Milvus**, supporting semantic retrieval across **1M+ embeddings** and reducing query response times to under a second for internal knowledge discovery tools.

Software Development Intern

May 2025 – Aug 2025

National Informatics Centre

Chennai, Tamil Nadu

- Built a **Spring Boot** ticketing platform integrated with **FastAPI** and **Qwen2.5-VL** to automate extraction of 8+ metadata fields, reducing ticket submission effort by **60–70%**.
- Developed a **RAG-based** support assistant with PostgreSQL integration across **10+ ticket categories** and workflows spanning **20+ states**, accelerating issue resolution for support teams.

PROJECTS

Printing Shop Management App (*GitHub*) | *Java, Spring Boot, Flutter, Cloudflare R2*

Oct 2025 – Jan 2026

- Built a Flutter application with **Spring Boot REST APIs** for document uploads, order processing, and real-time tracking across a **5,000+ student** campus environment.
- Integrated **Razorpay**, **Firebase**, and **Cloudflare R2** into a scalable workflow supporting high-volume uploads, automated payments, and notifications.

BinarySync | *Java, Spring Boot, AWS S3*

May 2026 – Jun 2026

- Built a cloud-native file versioning system using **content-defined chunking (CDC)** and **SHA-256 deduplication** to upload only modified portions of large binary files.
- Implemented concurrent S3 transfers using **Java 21 Virtual Threads** and manifest-based reconstruction through Spring Boot REST APIs.

NeuroScreen (*GitHub*) | *Python, FastAPI, Flutter, Machine Learning*

Feb 2026 – Jul 2026

- Developed a multimodal AI screening platform integrating speech, physical activity, and caregiver assessments.
- Built confidence scoring, progress tracking, and personalized therapy recommendations.

ACHIEVEMENTS, PROGRAMS AND CERTIFICATIONS

Hackathon Achievements (2026): Secured **1st Place** at BNY Service Design Jam, AI Wars, and Tekhora'26; **2nd Place** at VYUHATHON 1.0; **3rd Place** at DEV ARENA'26; and participated in **India Innovates 2026**, one of the world's largest civic-tech hackathons. (*Drive*)

AI Azure Internship Program (Microsoft Initiative) (May–Jun 2025): Built an ML-based AI vs Human text detection system during the AICTE–Edunet program. (*Drive*)

AWS Academy Cloud Foundations Graduate (Apr 2024) — Hands-on experience with core AWS services and cloud fundamentals. (*Drive*)